Software Engineering Lab 2

Agile Backlog Creation & Sprint Simulation in Jira

Community Solar Credit Allocation Manager

1. Project Overview

The Jira simulation converted the functional requirements of the Community Solar Credit Allocation Manager into Agile Epics and User Stories, assigned priorities and Fibonacci story points, and organized the backlog into two one-week sprints.

2. Agile Backlog

Epic	User Story	Priority	SP
Solar Generation & Credit Management	Record Solar Generation	High	8
Solar Generation & Credit Management	View Solar Credit Balance	High	3
Solar Generation & Credit Management	View Billing Offset	Medium	3
Solar Credit Transfer	Transfer Solar Credits	High	5
Solar Credit Transfer	Validate Credit Balance	High	3
Monitoring & Billing	Monitor Solar Activity	High	8
Monitoring & Billing	Generate Monthly Billing Statement	Medium	5
Monitoring & Billing	Apply Billing Offset	Medium	5

Total backlog: 40 story points.

3. Sprint Plan and Execution

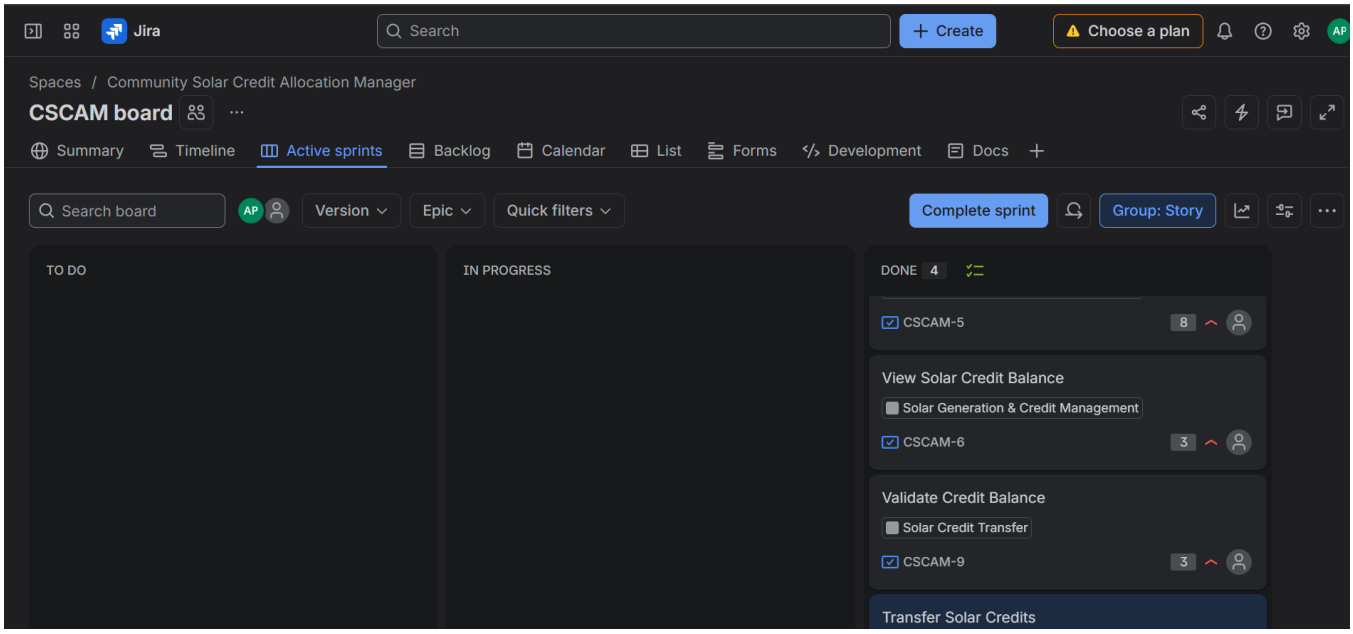
Sprint 1 — Solar Credit Foundation: Record Solar Generation (8), View Solar Credit Balance (3), Validate Credit Balance (3), and Transfer Solar Credits (5). Total: 19 story points. All planned stories were moved through In Progress to Done.

Sprint 2 — Monitoring & Billing: View Billing Offset (3), Monitor Solar Activity (8), Generate Monthly Billing Statement (5), and Apply Billing Offset (5). Total: 21 story points. All planned stories were moved through In Progress to Done.

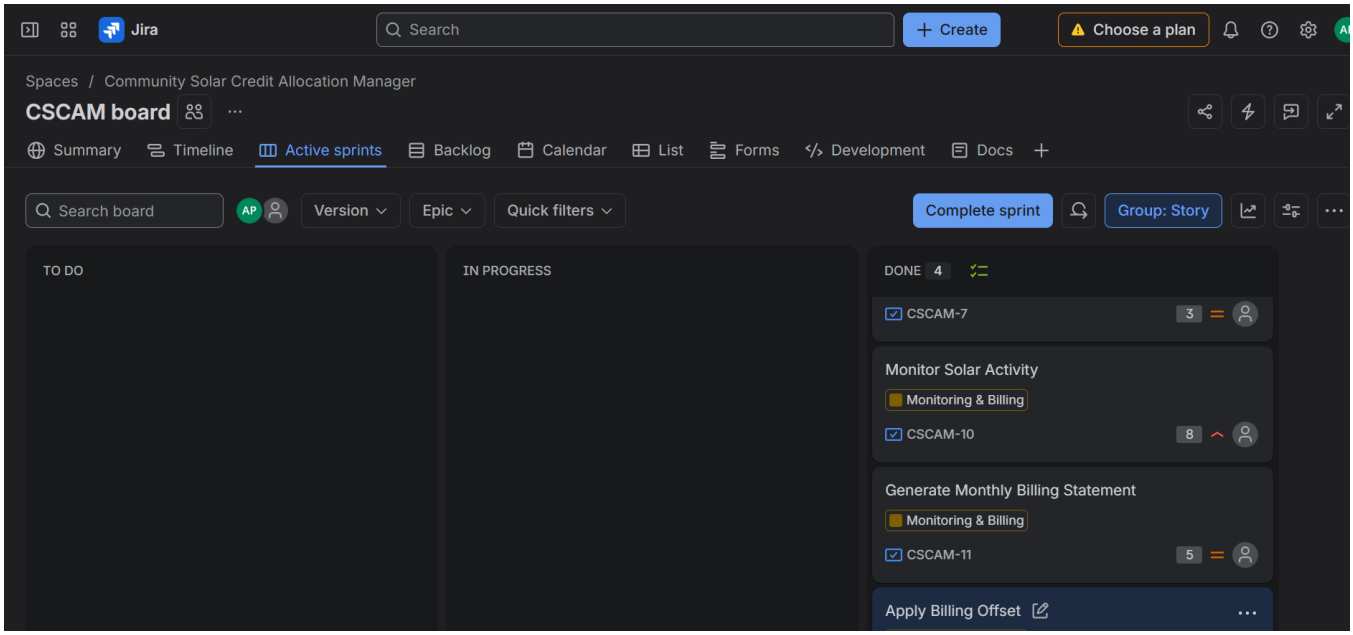
4. Jira Evidence

The following screenshots document the completed sprint boards and Sprint 2 burndown chart.

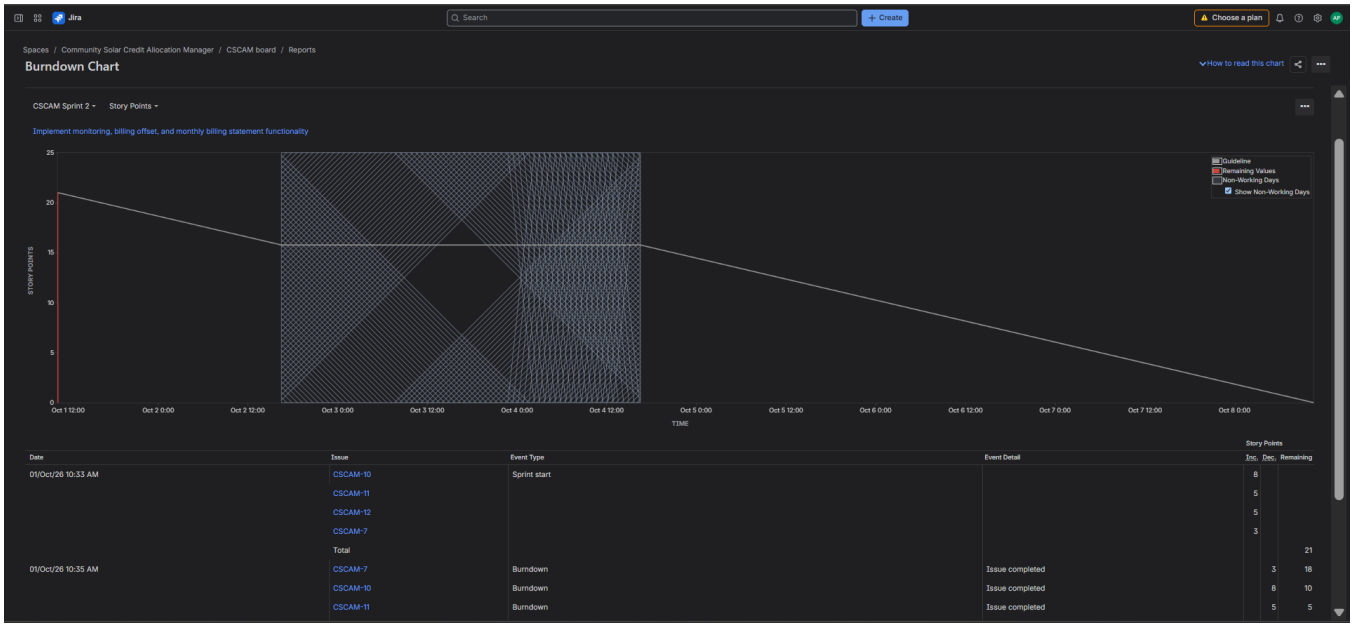
Jira Screenshot 1



Jira Screenshot 2



Jira Screenshot 3



Date	Issue	Event Type	Event Detail	Story Points Inc.	Dec.	Remaining
01/Oct/26 10:33 AM	CSCAM-10	Sprint start		8		
	CSCAM-11			5		
	CSCAM-12			5		
	CSCAM-7			3		
	Total					21
01/Oct/26 10:35 AM	CSCAM-7	Burndown	Issue completed		3	18
	CSCAM-10	Burndown	Issue completed		8	10
	CSCAM-11	Burndown	Issue completed		5	5

5. Burndown Chart Interpretation

Sprint 2 began with 21 story points. The Jira event table records completion of the 3-point, 8-point, and 5-point stories, reducing the remaining work from 21 to 18, then 10, then 5 points. The completed sprint ultimately reaches 0 remaining story points after the final 5-point story. The chart therefore records completion of all 21 planned Sprint 2 story points.

6. Reflection

Did your estimations reflect the actual effort?

The Fibonacci story-point estimates provided a reasonable representation of the relative effort of the user stories. Larger stories such as recording solar generation and monitoring solar activity were assigned 8 points, while smaller tasks such as viewing billing offsets were assigned 3 points. During the simulated sprints, all planned work was completed.

Was your backlog well-prioritized?

The backlog was prioritized according to the importance of the functionality. Core solar-generation, credit-balance, validation, and transfer functionality was given higher priority, while monitoring and billing features were scheduled in the second sprint according to the planned backlog structure.

How did your simulated sprint align with your plan?

The simulated sprints aligned with the planned workload. Sprint 1 contained 19 story points and Sprint 2 contained 21 story points. All planned stories in both sprints were moved through In Progress to Done, completing the 40-point backlog.

What insights did the burndown chart give about your team’s capacity?

The Sprint 2 burndown chart started at 21 story points and decreased to 0 as the four planned stories were completed. This demonstrated completion of the planned Sprint 2 workload. The burndown occurred in discrete steps corresponding to individual stories being completed rather than as a continuous decline.

7. Conclusion

The Jira exercise demonstrated backlog creation, Epic and User Story organization, prioritization, Fibonacci story-point estimation, sprint planning, sprint execution, and burndown analysis for the Community Solar Credit Allocation Manager.